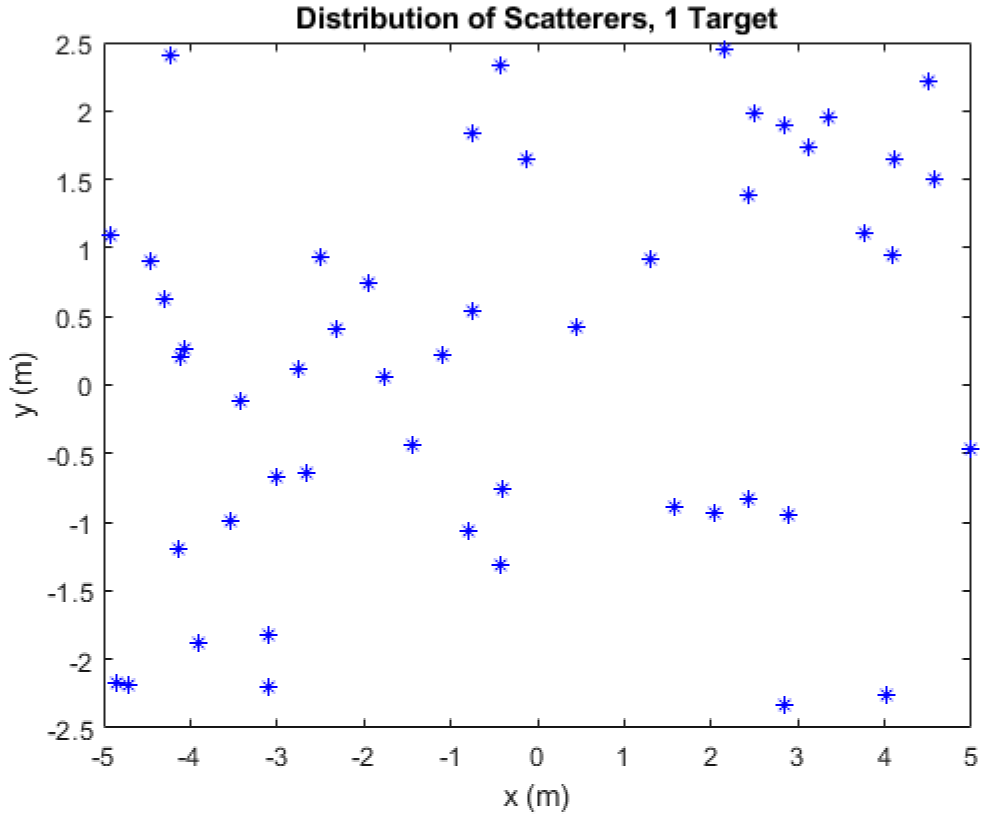
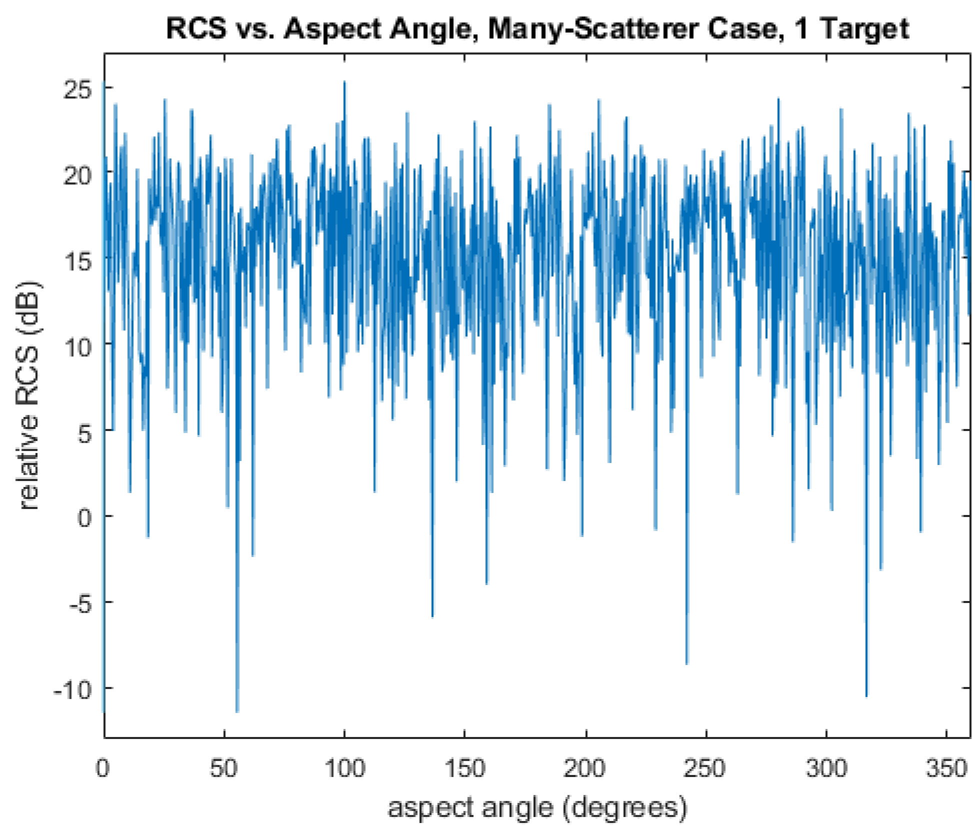
```
close all
clear
clc

Nt = 10; % number of targets
N = 50; %number of scatteres
z = 1; % RCS of each individual scatter sigma_i = 1 m^2

% 50 point scatters uniformly random distributed in a box +- 5 meters, and
% +- 2.5 meters in y dimension centered at (x,y) = (0,0), for 10 different
% targets

figure(1)
x = 10*rand(N,Nt)-5;
y = 5*rand(N,Nt)-2.5;
plot(x(:,1),y(:,1),'*b')
axis([-5 5 -2.5 2.5]);
xlabel('x (m)');
ylabel('y (m)');
title('Distribution of Scatterers, 1 Target');
saveas(gcf,'Distribution of Scatterers, 1 Target.png');

% Input number of angles and nominal range and frequency
% M=input('Enter # of angles: ');
% R=input('Enter nominal range (m): ');
% f=input('Enter frequency (Hz): ');
M = round(360/0.5);
R = 10e3;
f = 10e9;

% Loop over aspect angle to do complex voltage estimates
q = zeros(M);
echo=zeros(M,Nt);

% Loop over radar-target aspect angles

for k=1:M
    q(k)=(2*pi/M)*(k-1); % current aspect angle

    % Loop over targets and individual point scatterers
    % each column of 'echo' is a different target

    for r = 1:Nt
        for p=1:N
            phasor=z*exp(j*4*pi*f*norm([x(p,r)-R*cos(q(k)),y(p,r)-
R*sin(q(k))])/3e8);
            echo(k,r)=echo(k,r)+phasor;
        end
    end
end

% The magnitude-squared of the total complex echo
```

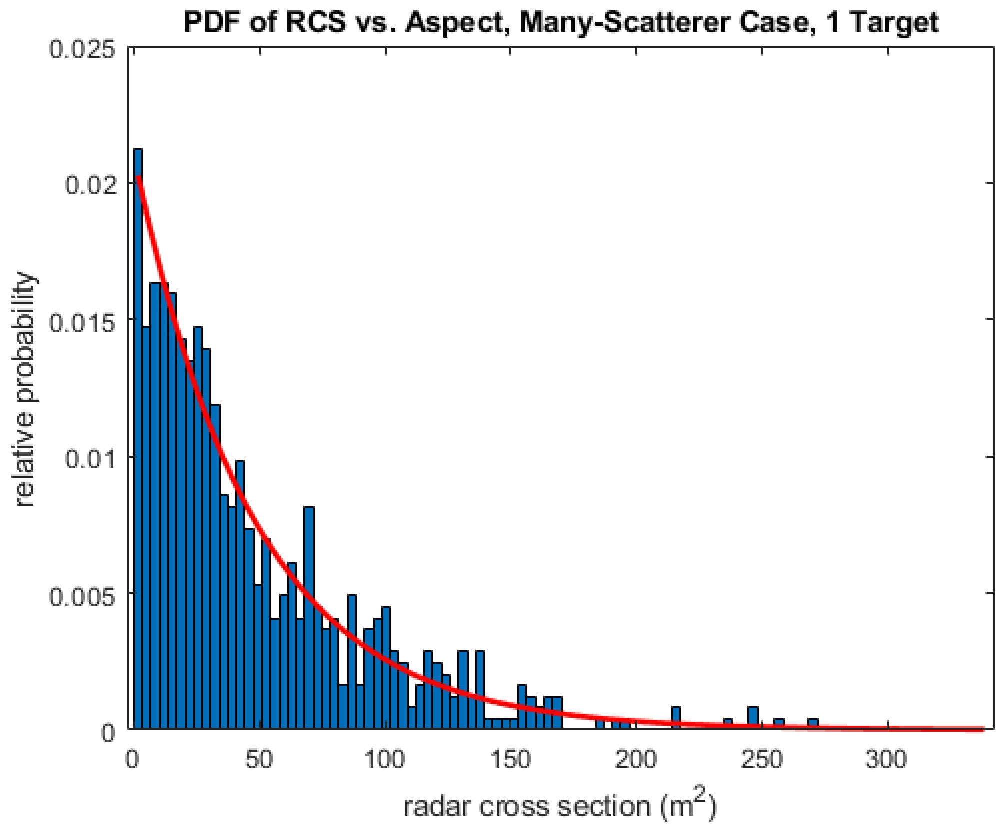
```matlab
% is the RCS to within a constant

RCS = abs(echo).^2;
RCSdB = 10*log10(RCS);

% Plot the dB data for the first target

figure(2)
plot((180/pi)*q,RCSdB(:,1));
xlabel('aspect angle (degrees)');
ylabel('relative RCS (dB)');
title('RCS vs. Aspect Angle, Many-Scatterer Case, 1 Target')
axis([0,360,10*log10(N*z)-30,10*log10(N*z)+10]);
saveas(gcf,'RCS vs. Aspect Angle, Many-Scatterer Case, 1 Target.png');

% Compute a 100-bin histogram of the data for each target;
% plot the theoretical exponential distribution as well

density = zeros(100,Nt);
centers = [];
for r = 1:Nt
    if (isempty(centers))
        [count,centers]=hist(RCS(:,r),100);
    else
        [count,centers]=hist(RCS(:,r),centers);
    end
    bin_size = centers(2) - centers(1);
    area = sum(count)*bin_size;
    density(:,r) = count/area;
    figure(3)
    bar(centers,density(:,r),1);
    hold on
    msig = mean(RCS(:,r)); % compute mean of current linear-scale RCS data
    exp_pdf = exp(-centers/msig)/msig; % theoretical pdf on same x-axis values

    plot(centers,exp_pdf,'r','LineWidth',2)
    xlabel('radar cross section (m^2)')
    ylabel('relative probability')
    title('PDF of RCS vs. Aspect, Many-Scatterer Case, 1 Target')
    hold off
    pause(0.5)
end
saveas(gcf,'PDF of RCS vs. Aspect, Many-Scatterer Case, 1 Target.png');

% Now average the separate histograms and plot with an overlaid exponential
% based on the mean of all the data for all of the targets

figure(4);
bar(centers,mean(density,2),1);
hold on
msig = mean(RCS(:));  % compute mean of all linear-scale RCS data
exp_pdf = exp(-centers/msig)/msig;  % theoretical pdf on same x-axis values
plot(centers,exp_pdf,'r','LineWidth',2)
xlabel('radar cross section (m^2)')
```

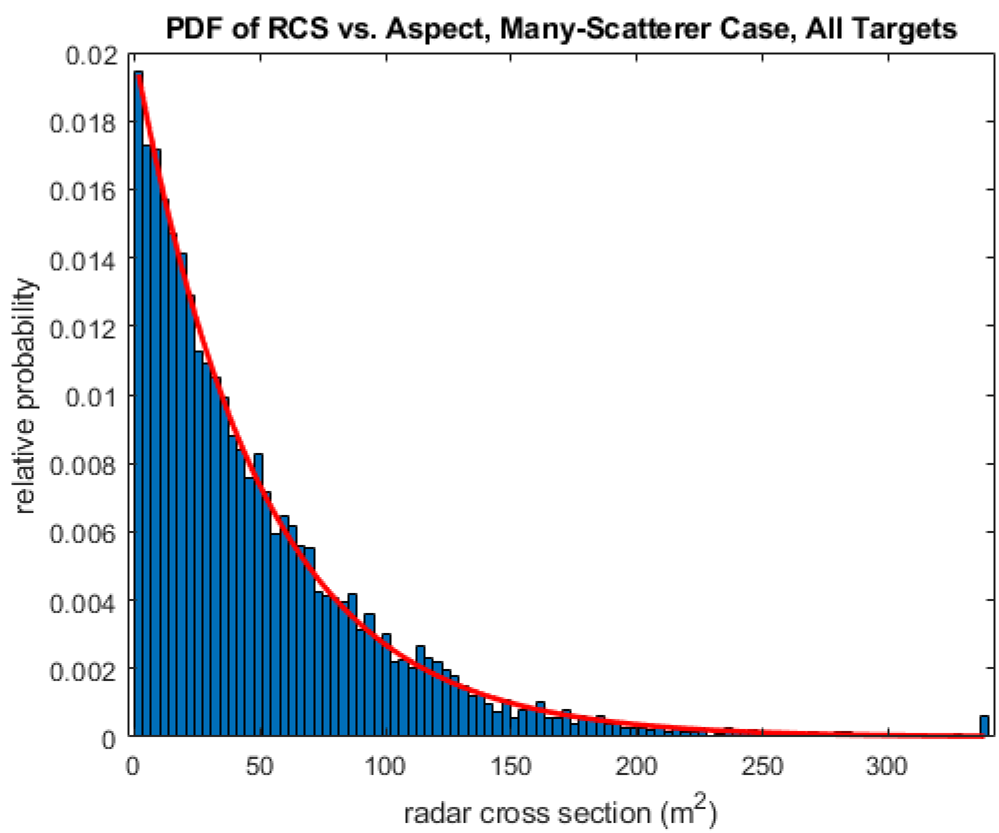
```
ylabel('relative probability')
title('PDF of RCS vs. Aspect, Many-Scatterer Case, All Targets')
hold off
saveas(gcf,'PDF of RCS vs. Aspect, Many-Scatterer Case, All Targets.png');

% Now do the variation with frequency

% Input number of angles and nominal range and frequency
% Mf=input('Enter # of frequencies: ');
% f1=input('Enter start frequency (Hz): ');
% df=input('Enter freq step (Hz)');

Mf = 300;
f1 = 10e9;
df = 15e6;
fv = f1+(0:Mf-1)*df;

% Loop over radar frequencies

echof=zeros(Mf,Nt);
for k=1:Mf
    % Loop over targets and individual point scatterers
    % each column of 'echof' is a different target
    for r = 1:Nt
        for p=1:N
            phasor=z*exp(j*4*pi*fv(k)*norm([x(p,r)-R,y(p,r)])/3e8);
            echof(k,r)=echof(k,r)+phasor;
        end
    end
end

% The magnitude-squared of the total complex echo
% is the RCS to within a constant

RCSf=abs(echof).^2;
RCSfdB=10*log10(RCSf);

% Plot the dB data for the first target as an example

figure(5)
plot(fv,RCSfdB(:,1)); xlabel('RF frequency (Hz)');
ylabel('relative RCS (dB)');
title('RCS vs. RF Frequency, Many-Scatterer Case, 1 Target')
axis([10e9,14.5e9,10*log10(N*z)-30,10*log10(N*z)+10]);
saveas(gcf,'RCS vs. RF Frequency, Many-Scatterer Case, 1 Target.png');


% Compute a 100-bin histogram of the data for each target;
% plot the theoretical exponential distribution as well

densityf = zeros(100,Nt);
centersf = [];
for r = 1:Nt
    if (isempty(centersf))
```

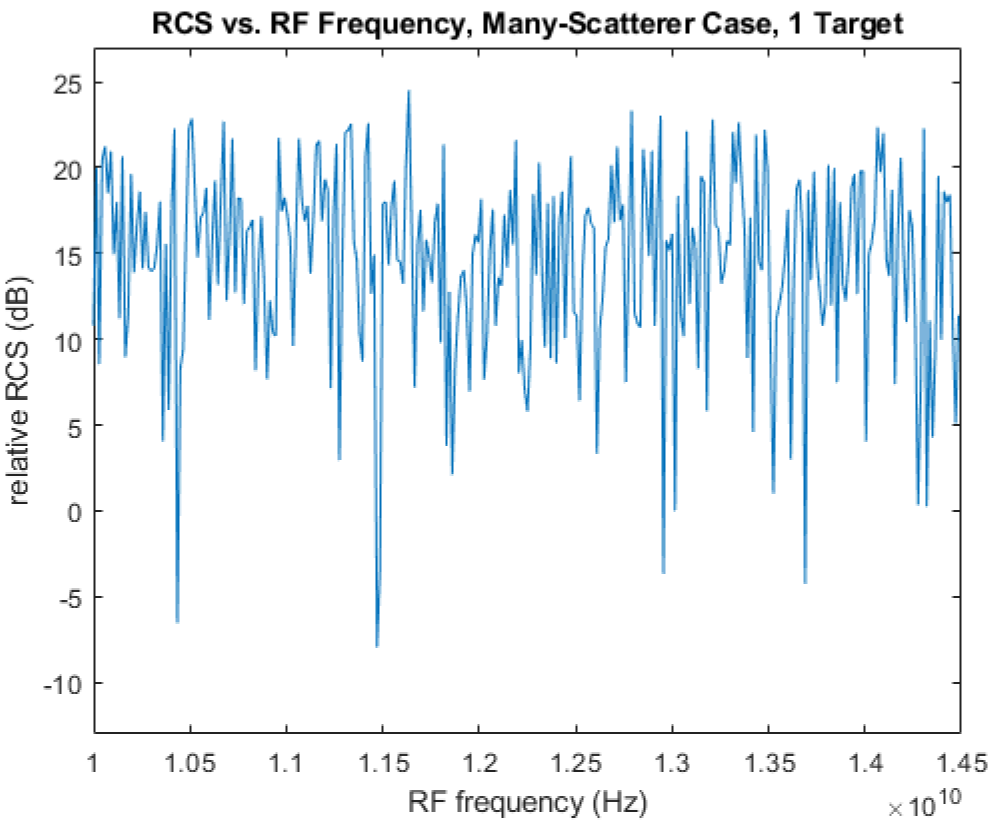
```
        [countf,centersf]=hist(RCSf(:,r),100);
    else
        [countf,centersf]=hist(RCSf(:,r),centersf);
    end
    bin_sizef = centersf(2)-centersf(1);
    areaf = sum(countf)*bin_sizef;
    densityf(:,r) = countf/areaf;
    % Plot the histogram of the data for a single target, and an overlaid
    % exponential for that target, based only on that target's data
        figure(6);
        bar(centersf,densityf(:,r),1);
        hold on
        msigf=mean(RCSf(:,r)); % compute mean of current linear-scale RCS data
        exp_pdf = exp(-centersf/msigf)/msigf; % theoretical pdf on same x-axis
 values

        plot(centersf,exp_pdf,'r','LineWidth',2)
        xlabel('radar cross section (m^2)')
        ylabel('relative probability')
        title('PDF of RCS vs. RF, Many-Scatterer Case, 1 Target')
        hold off
        pause(0.5)
end
saveas(gcf,'PDF of RCS vs. RF, Many-Scatterer Case, 1 Target.png');

% Now average the separate histograms and plot with an overlaid exponential
% based on the mean of all the data for all of the targets

figure(7);
bar(centersf,mean(densityf,2),1);
hold on
msigf=mean(RCSf(:));  % compute mean of all linear-scale RCS data
exp_pdf = exp(-centersf/msigf)/msigf;  % theoretical pdf on same x-axis values
plot(centersf,exp_pdf,'r','LineWidth',2)
xlabel('radar cross section (m^2)')
ylabel('relative probability')
title('PDF of RCS vs. RF, Many-Scatterer Case, All Targets')
hold off
saveas(gcf,'PDF of RCS vs. RF, Many-Scatterer Case, All Targets.png');
```

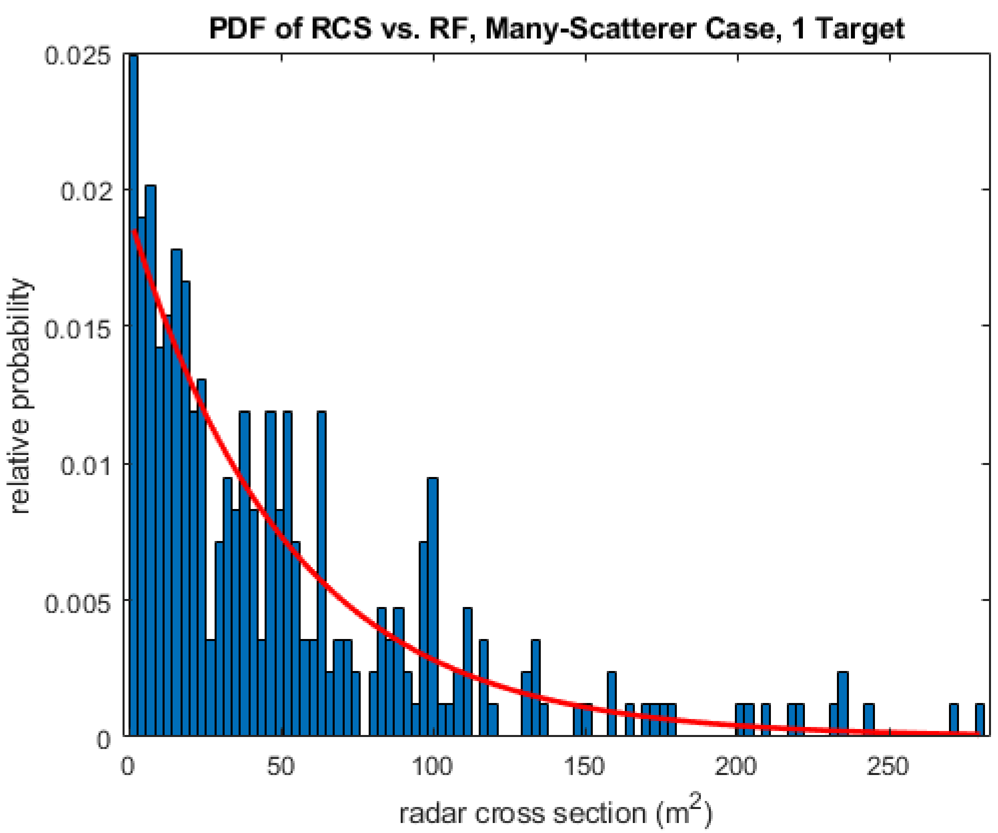

Distribution of Scatterers, 1 Target
y (m)
x (m)
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
-5
-4
-3
-2
-1
0
1
2
3
4
5

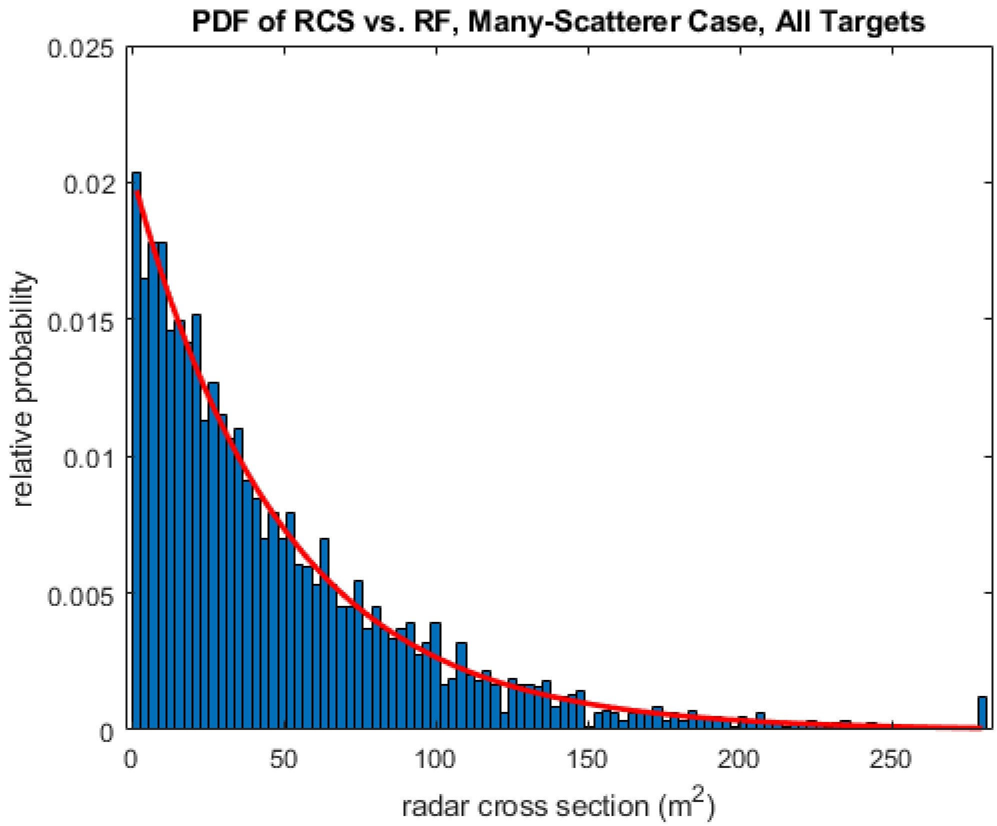

RCS vs. Aspect Angle, Many-Scatterer Case, 1 Target
relative RCS (dB)
aspect angle (degrees)
25
20
15
10
5
0
-5
-10
0
50
100
150
200
250
300
350


PDF of RCS vs. Aspect, Many-Scatterer Case, 1 Target
relative probability
0.025
0.02
0.015
0.01
0.005
0
0
50
100
150
200
250
300
radar cross section (m2)


PDF of RCS vs. Aspect, Many-Scatterer Case, All Targets
relative probability
0.02
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
0
50
100
150
200
250
300
radar cross section (m2)


RCS vs. RF Frequency, Many-Scatterer Case, 1 Target
relative RCS (dB)
25
20
15
10
5
0
-5
-10
1
1.05
1.1
1.15
1.2
1.25
1.3
1.35
1.4
1.45
RF frequency (Hz)
× 10^10


PDF of RCS vs. RF, Many-Scatterer Case, 1 Target
relative probability
0.025
0.02
0.015
0.01
0.005
0
0
50
100
150
200
250
radar cross section (m^2)



Published with MATLAB® R2022b